Training Humanoid Robot on Simulators and Causal World Model

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Research Background

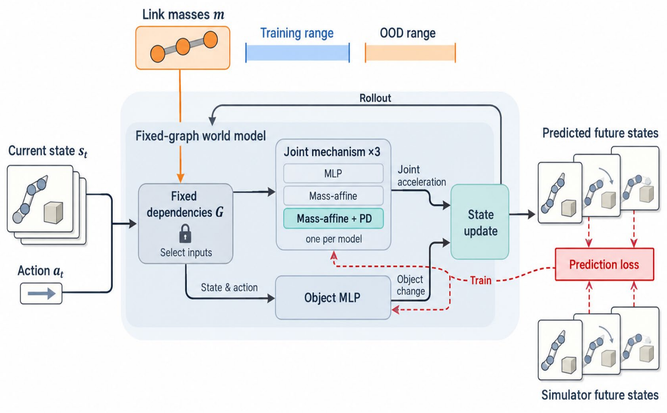
- Embodied AI integrated into robotics is set to revolutionize diverse sectors, ranging from industrial manufacturing and public services to healthcare facilities and domestic care.
- Training humanoid robot on simulators and causal world model is imperative in embodied AI development.

Objective

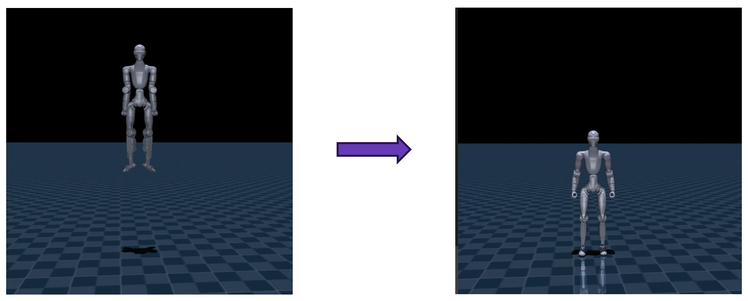
- Develop methods to train humanoid robots in locomotion and functional tasks, such as standing, walking, opening doors, item retrieval.
- Deploy trained models onto a physical Unitree R1 EDU.
- Different approaches to autonomous control via artificial intelligence or reinforcement learning to explore the profound impact of the body's interactions with the environment on cognitive processes in a causal world model.

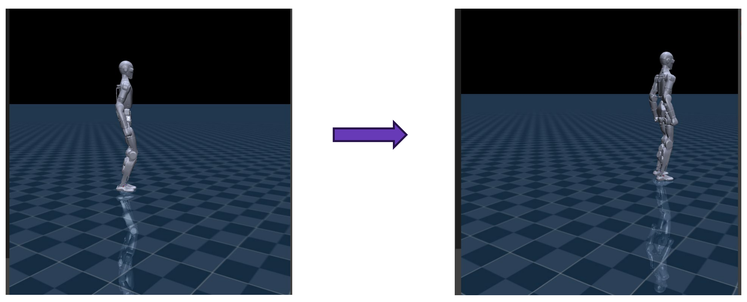
Causal World Model Approach

- Developed accurate robot joint-dynamics models that generalize to unfamiliar link masses.
- **Models compared:** standard MLP, mass-affine dynamics with learned torque, and mass-affine dynamics with known PD control.
- **Model Design:** structured models explicitly incorporate link masses into inertia and bias terms, while neural networks learn state-dependent dynamics.
- **World Model:** predicts joint motion and object-state changes over time using numerical integration and separate MLPs.
- **Training:** Tested on separate link-mass ranges, including 150-step trajectory rollouts.
- **Mass-information ablation:** Compared models using actual masses against controls that removed or fixed mass inputs.
- **Results:** Under unfamiliar link masses, the mass-affine model with PD control achieved a median time average joint R2 of 0.727, compared with 0.609 when mass inputs were fixed at their training means.



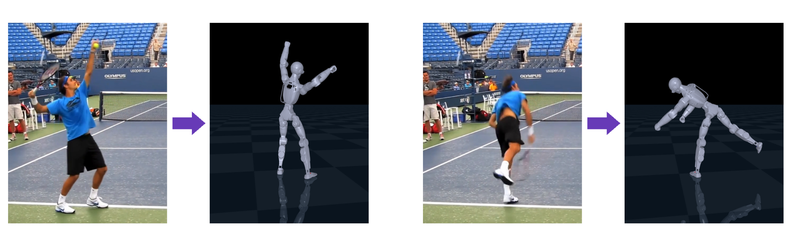
Gantry Supported RL Development





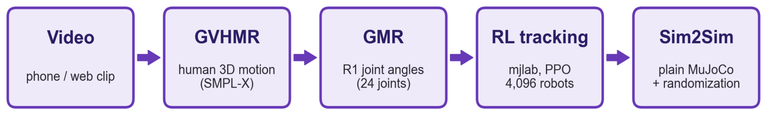
- Set up Unitree's R1 locomotion Sim2Sim pipeline using MJLab/MuJoCo with the deployment controller, ONNX policy, and Unitree SDK2 interface.
- Validated the full locomotion path from simulated LowState → policy inference → LowCmd motor commands, including joint mapping and command behavior.
- Developed a gantry-supported simulation test environment to mimic planned real-world suspended/assisted R1 bring-up and lowering tests.
- Performed initial Sim2Real safety/readiness checks, including monitoring commanded joint ranges and confirming no simulated joint-limit violations during walking.

Video-to-R1 Motion Imitation (MJLAB)



Ball toss **Follow-through**

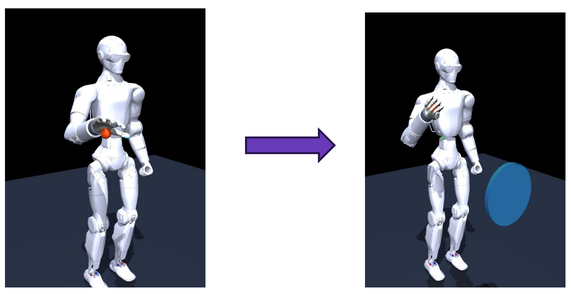
Original video (left) → same motion by the R1 policy in MuJoCo (right). Video: GVHMR example clip.



- ❑ Built a **video → R1 pipeline**: GVHMR recovers the human's 3D motion, **GMR** retargets it onto the R1's 24 joints, and a **tracking policy** is trained in mjlab.
- ❑ **Sim2Sim check** in plain MuJoCo: the R1 completes the full serve in **30/30** randomized runs.
- ❑ **Sim2Real risk found:** a **40 ms command delay** drops survival to 2/20; training with random delay restores **15/20**.

Arm-Hand MUJOCO Training

- Used MuJoCo to simulate the Unitree R1 humanoid arm with a BrainCo Revo 2 Basic robotic hand.
- The task is to reach for a ball, grasp it, maintain the grip during windup, and throw it toward a target.
- **Control:** neural policy controlling 5 arm joints and 6 active hand joints.
- **Training:** 63,985 state-action samples from 240 demonstrations, using behavioral cloning and 2 DAgger correction rounds.
- **Domain Randomization:** varied ball properties, friction, actuator gains, observation noise, command delay, and target location.
- **Performance:** across 100 randomized trials, 85% complete-task success, 99% reach success, 96 % grasp success, and 91 % hold-and-release success, 13.99 mm average reach error, 0.209 m average throw distance.



Conclusions

- In the CausalWorld benchmark, the learned-torque setting is competitive with the strongest tested RWM configuration.
- Based on the gantry supported RL locomotion task, we can now integrate the model onto the physical Unitree R1.
- Video to R1 motion imitation (MJLAB) pipeline for training is accomplished.
- The neural policy achieved reliable simulated reach-grasp-throw behavior, reaching 85% complete-task success.

Acknowledgment

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References

[1] CausalWorld contributors. TriFingerRobot: PD control and safety torque implementation. CausalWorld source documentation, version 1.2.0. Accessed October 7, 2026.

[2] O. Ahmed, F. Träuble, A. Goyal, A. Neitz, Y. Bengio, B. Schölkopf, M. Wüthrich, and S. Bauer. CausalWorld: A Robotic Manipulation Benchmark for Causal Structure and Transfer Learning. ICLR, 2021.